# ACTIVITY 1: LINEAR REGRESSION USING THEOPH DATASET

## 1.1 LOAD AND INSPECT DATASET

```
> head(Theoph)
  Subject   Wt Dose Time  conc
1       1 79.6 4.02 0.00  0.74
2       1 79.6 4.02 0.25  2.84
3       1 79.6 4.02 0.57  6.57
4       1 79.6 4.02 1.12 10.50
5       1 79.6 4.02 2.02  9.66
6       1 79.6 4.02 3.82  8.58
> str(Theoph)
Classes ‘nfnGroupedData’, ‘nfGroupedData’, ‘groupedData’ and 'data.frame':      132 obs. of  5 variables:
 $ Subject: Ord.factor w/ 12 levels "6"<"7"<"8"<"11"<..: 11 11 11 11 11 11 11 11 11 11 ...
 $ Wt     : num  79.6 79.6 79.6 79.6 79.6 79.6 79.6 79.6 79.6 79.6 ...
 $ Dose   : num  4.02 4.02 4.02 4.02 4.02 4.02 4.02 4.02 4.02 4.02 ...
 $ Time   : num  0 0.25 0.57 1.12 2.02 ...
 $ conc   : num  0.74 2.84 6.57 10.5 9.66 8.58 8.36 7.47 6.89 5.94 ...
 - attr(*, "formula")=Class 'formula'  language conc ~ Time | Subject
  .. ..- attr(*, ".Environment")=<environment: R_EmptyEnv>
 - attr(*, "labels")=List of 2
  ..$ x: chr "Time since drug administration"
  ..$ y: chr "Theophylline concentration in serum"
 - attr(*, "units")=List of 2
  ..$ x: chr "(hr)"
  ..$ y: chr "(mg/l)"
```

The Theoph dataset was loaded and inspected using head() and str(). The variables Wt (weight) and Dose were identified and used for the linear regression model.

## 1.2 CREATE LINEAR REGRESSION MODEL

```
> model_theoph <- lm(Dose ~ Wt, data = Theoph)
> print(model_theoph)

Call:
lm(formula = Dose ~ Wt, data = Theoph)

Coefficients:
(Intercept)           Wt
   10.03988     -0.07781
```

A linear regression model was created using Dose as the dependent variable and Wt as the independent variable.

## 1.3 MODEL SUMMARY

```
> summary(model_theoph)

Call:
lm(formula = Dose ~ Wt, data = Theoph)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.21738 -0.04004 -0.01802  0.02959  0.18465 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) 10.0398771  0.0695360  144.38   <2e-16 ***
Wt          -0.0778066  0.0009909  -78.52   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1036 on 130 degrees of freedom
Multiple R-squared:  0.9794,	Adjusted R-squared:  0.9792 
F-statistic:  6166 on 1 and 130 DF,  p-value: < 2.2e-16
```

**Discussion:**

The summary() function was used to evaluate the performance of the linear regression model. The output shows that the intercept is 10.0399, while the coefficient for Wt is -0.0778. This indicates that for every 1 kg increase in weight, the predicted dose decreases by approximately 0.0778 mg/kg, showing a negative relationship between weight and dose.

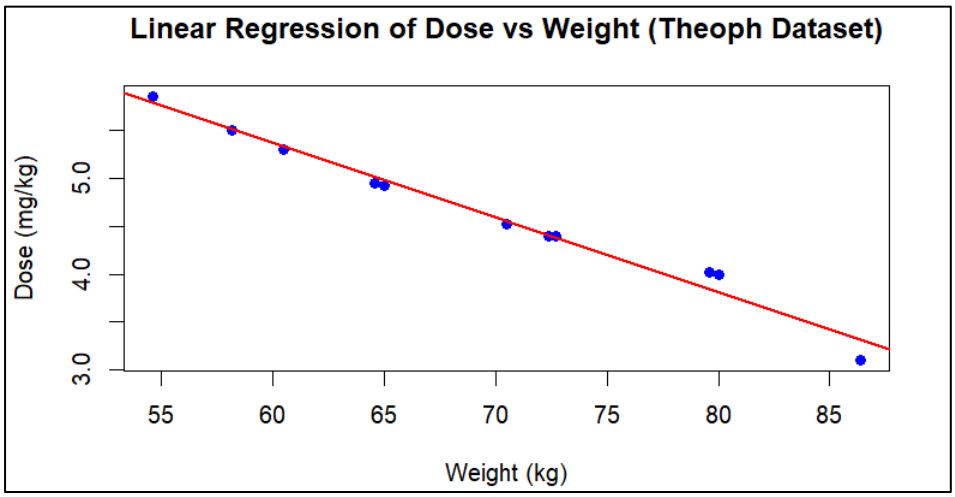
The p-value for the variable Wt is less than 2e-16, which is significantly smaller than 0.05. This means that weight is a statistically significant predictor of dose, and there is strong evidence to support the relationship between the two variables.

The model produced a Multiple R-squared value of 0.9794, which means that 97.94% of the variation in dose can be explained by the weight variable. This indicates that the model has a very strong fit to the data. The Adjusted R-squared value of 0.9792 is very close to the R-squared value, confirming that the model remains reliable even after adjustment.

Additionally, the F-statistic is 6166 with a p-value less than 2.2e-16, which shows that the overall regression model is highly significant.

Overall, the summary results indicate that there is a strong and statistically significant negative linear relationship between weight and dose in the Theoph dataset.

## 1.4 VISUALISATION




The scatter plot with regression line shows a negative linear relationship between weight and dose.

## 1.5 PREDICTION

```
> new_weight <- data.frame(Wt = c(90, 95, 100))
> predicted_dose <- predict(model_theoph, newdata = new_weight)
>
> print(new_weight)
   Wt
1  90
2  95
3 100
> print(predicted_dose)
       1        2        3
3.037282 2.648249 2.259215
```

**Explanation:**

The predict() function was used to estimate the dose for new weight values of 90 kg, 95 kg, and 100 kg. A new data frame was created to store these weight values, and the fitted linear regression model was applied to generate the predicted doses. The predicted results are:

- 90 kg → 3.037 mg/kg
- 95 kg → 2.648 mg/kg
- 100 kg → 2.259 mg/kg

These results show a decreasing trend in dose as weight increases, which is consistent with the negative coefficient obtained from the regression model. This confirms that there is an inverse relationship between weight and dose.

It should also be noted that these weight values are slightly outside the main range of the dataset. Therefore, the predictions are based on extrapolation and may be less accurate compared to predictions within the observed data range.

## ACTIVITY 2: K-NEAREST NEIGHBOR (KNN) USING ChickWeight DATASET

The Diet variable was converted into a factor to represent categorical class labels required for classification. Only the relevant variables (Time, weight, and Diet) were selected for the analysis.

The dataset was split into training and testing sets using a 70:30 ratio. The predictor variables were scaled using the scale() function to ensure fair distance calculation in the KNN algorithm.

Several values of K were tested (1, 3, 5, 7, 9, 11, 13, and 15), and the classification accuracy was calculated for each. Based on the results, **K = 9** produced the highest accuracy of **0.4104** and was selected as the optimal K value.

The final KNN model was then built using K = 9. The confusion matrix shows the comparison between actual and predicted diet classes. The diagonal values represent correct classifications, while the off-diagonal values represent misclassifications.

The overall model accuracy is **0.4104**, indicating that approximately 41.04% of the test data was classified correctly. This relatively low accuracy suggests that the features Time and weight alone may not be sufficient to clearly distinguish between different diet groups.

```
> print(accuracy_results)
   K  Accuracy
1  1 0.3236994
2  3 0.3815029
3  5 0.3872832
4  7 0.3815029
5  9 0.4104046
6 11 0.3930636
7 13 0.3988439
8 15 0.4046243
```

*Figure 1: Accuracy results for different K values*

```
> print(cm)
      Predicted
Actual  1  2  3  4
     1 51  6  5  4
     2 27  4  1  4
     3 12  7  5 12
     4 14  4  6 11
>
> final_accuracy <- mean(final_pred == test_y)
> print(final_accuracy)
[1] 0.4104046
```

*Figure 2: Confusion matrix and final accuracy of the KNN model*